

# Gradient and Durbin Tests for Target Detection in Structural Nonhomogeneity Arising from the Generalized Eigenrelation

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**Abstract**—This paper tackles the challenge of adaptive target detection for subspace signals in the case of the structural nonhomogeneity. A typical kind of structural nonhomogeneity caused by the generalized eigenrelation is considered, and then we employ the gradient and Durbin tests for detector design. Our analysis demonstrates that the gradient test is coincident with the existing detector based upon the generalized likelihood ratio test, whereas the proposed Durbin test is new. Simulation experiments assess the detection capability of the Durbin test, and it performs well in the presence of a slight structural nonhomogeneity.

**Index Terms**—Adaptive detection, structure nonhomogeneous environment, gradient test, Durbin test, generalized eigenrelation, subspace signal.

## I. INTRODUCTION

Multichannel adaptive signal detection has received extensive attention and attracted substantial research interest in recent decades [1]. Kelly [2] first addressed multichannel adaptive signal detection for rank-one signals, and developed the detector by resorting to the generalized likelihood ratio test (GLRT) criterion. A rank-one signal implies that the direction vector of a signal is completely known, while the signal's amplitude is unknown. Subsequently, the Wald test [3] and Rao test [4] were successively derived. These above three criteria serve as common detection design criteria. As an extension model of a rank-one signal, the subspace signal is applicable to a wider range of scenarios. The subspace signal model presumes that a signal resides within a specified subspace, while its coordinates are unknown. In recent decades, numerous subspace-based detectors have been developed as summarized in [1].

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Most of the aforementioned references focus on homogeneous environment. However, the real scenarios often involve various uncertainties and unfavorable factors, frequently leading to environmental nonhomogeneity, including two main types: power nonhomogeneity and structural nonhomogeneity [5]. The most common and typical model of power nonhomogeneity is partial homogeneity, i.e., a scenario in which the primary data and secondary data exhibit an identical covariance matrix (CM) structure but differ only by an unknown power factor. In contrast, structural nonhomogeneity means the CM structures of the primary and secondary data are inconsistent. For partial homogeneity, various subspace-based detectors were designed in [6]. Building upon this foundation, [7] further considered the presence of interference, and the detection problem was handled according to the above three criteria, gradient test and Durbin test. It is noteworthy that the gradient and Durbin tests have also become popular detector design criteria in recent years [8]–[10]. In structure nonhomogeneous environment (SNE), a kind of structural nonhomogeneity assumes that the random CMs of primary and secondary data are governed by a given joint distribution [11]. Reference [5] considers another kind of structural nonhomogeneity, which is to describe the mismatch of the CMs by an unknown random jammer. In addition, [12] assumes that the unknown random jammer causing the CM mismatch is modeled under the constraint of the generalized eigenrelation (GER). Subsequently, three subspace-based detectors according to the GLRT, Rao test, and Wald test were derived in [13].

For adaptive subspace signal detection in SNE caused by the GER, it is noteworthy to highlight the absence of uniformly most powerful test. Moreover, the gradient and Durbin tests remain unexplored up to this point. Therefore, this paper reconsiders the adaptive detection problem for a subspace signal in SNE arising from the GER. We employ the gradient

and Durbin tests for detector design, and analysis results imply that the gradient test is coincident with the existing GLRT-based counterpart, whereas the Durbin test presents a new detector. Simulation experiments demonstrate that the proposed Durbin test is capable of effectively detecting the target, and it exhibits comparable performance to that of the GLRT and Wald test with chosen parameters. When there is a slight structural nonhomogeneity, the Durbin test also performs well.

## II. PROBLEM FORMULATION

The primary data, described as an  $N \times 1$  vector  $\mathbf{z}$ , is collected from an array of  $N$  sensors. The binary hypothesis test is composed of hypothesis  $H_1$  that  $\mathbf{z}$  contains signal  $\mathbf{s}$  and noise  $\mathbf{n}$ , and hypothesis  $H_0$  that  $\mathbf{z}$  contains noise  $\mathbf{n}$  only. The signal  $\mathbf{s}$  is expressed as  $\mathbf{s} = \mathbf{D}\phi$ , where the  $N \times q$  matrix  $\mathbf{D}$  is known and full-column-rank, while  $\phi$  is a  $q \times 1$  unknown vector. The columns of  $\mathbf{D}$  spans the signal subspace, whereas  $\phi$  represents the coordinates within this subspace. The noise component  $\mathbf{n}$  is distributed as  $\mathcal{CN}_N(\mathbf{0}, \mathbf{R}_t)^1$ , and  $\mathbf{R}_t$  is generally not directly obtainable in practical applications. Therefore, a secondary data set  $\mathbf{z}_k, k = 1, \dots, K$  is commonly presumed to be available for estimating  $\mathbf{R}_t$ . The secondary data  $\mathbf{z}_k$  only contains noise  $\mathbf{n}_k$ , which is characterized by  $\mathcal{CN}_N(\mathbf{0}, \mathbf{R})$ .

For structural nonhomogeneity, we suppose that  $\mathbf{R}_t$  and  $\mathbf{R}$  satisfy the relationship  $\mathbf{R}_t = \mathbf{R} + \mathbf{p}\mathbf{p}^H$ , where the  $N \times 1$  unknown vector  $\mathbf{p}$  stands for the random jammer resulting in the structural mismatch between  $\mathbf{R}_t$  and  $\mathbf{R}$ . Moreover, it is assumed that  $\mathbf{p}$  is constrained by the GER, i.e., the unknown random jammer  $\mathbf{p}$  is constrained to [12]

$$\mathbf{D}^H \mathbf{R}^{-1} \mathbf{p} = \mathbf{0}_{q \times 1}. \quad (1)$$

It is worth pointing out that the GER often approximately holds true under some conditions, and can also be used to characterize the undernullified interference or unexpected discrete [12], [13].

To summarize, we consider the detection problem as follows:

$$\begin{cases} H_0 : \mathbf{z} = \mathbf{n}, & \mathbf{z}_k = \mathbf{n}_k, \quad k = 1, \dots, K, \\ H_1 : \mathbf{z} = \mathbf{D}\phi + \mathbf{n}, & \mathbf{z}_k = \mathbf{n}_k, \quad k = 1, \dots, K. \end{cases} \quad (2)$$

## III. DETECTOR DESIGN

The general expressions of the statistics for the gradient and Durbin tests can be written as [8]

$$t_{\text{Gradient}} = \left. \frac{\partial \ln f_1(\mathbf{z}, \mathbf{Z}_K)}{\partial \psi_r^T} \right|_{\psi = \hat{\psi}_0} (\hat{\psi}_{r1} - \psi_{r0}), \quad (3)$$

<sup>1</sup>The symbols  $(\cdot)^T$ ,  $(\cdot)^H$ ,  $|\cdot|$ ,  $\text{vec}(\cdot)$ ,  $\text{tr}(\cdot)$ , and  $\mathbb{E}[\cdot]$  stand for the transpose, conjugate transpose, determinant, vectorization, trace, and statistical expectation operators, respectively. The notations  $\mathbf{0}$  and  $\mathbf{I}$  represent a null matrix (vector) and identity matrix of proper size, respectively. The symbol  $\mathcal{CN}_N(\mathbf{r}, \mathbf{Q})$  stands for an  $N \times 1$  complex circular normal random distribution, characterized by a mean vector  $\mathbf{r}$  and a CM  $\mathbf{Q}$ . The orthogonal projection matrix onto the range space of  $\mathbf{B}$  is  $\mathbf{P}_B = \mathbf{B}(\mathbf{B}^H \mathbf{B})^{-1} \mathbf{B}^H$ , and  $\mathbf{P}_B^\perp = \mathbf{I} - \mathbf{P}_B$ .

and

$$t_{\text{Durbin}} = (\hat{\psi}_{r,01} - \psi_{r0})^H \left\{ \left[ \mathbf{F}(\hat{\psi}_0) \right]_{\psi_r, \psi_r} \left[ \mathbf{F}^{-1}(\hat{\psi}_0) \right]_{\psi_r, \psi_r} \right. \\ \left. \times \left[ \mathbf{F}(\hat{\psi}_0) \right]_{\psi_r, \psi_r} \right\} (\hat{\psi}_{r,01} - \psi_{r0}), \quad (4)$$

respectively, where  $\mathbf{Z}_K = [\mathbf{z}_1, \dots, \mathbf{z}_K]$ ,  $f_\lambda(\mathbf{z}, \mathbf{Z}_K), \lambda = 0, 1$  is the joint probability density functions (PDFs) of  $\mathbf{z}$  and  $\mathbf{Z}_K$  under  $H_\lambda$ ,  $\psi_r$ ,  $\psi_s$ , and  $\psi = [\psi_r^T, \psi_s^T]^T$  denote the unknown relative, nuisance, and all parameters, respectively,  $\psi_{r0}$  is the value of  $\psi_r$  under  $H_0$ ,  $\hat{\psi}_{r1}$ ,  $\hat{\psi}_{s0}$ , and  $\hat{\psi}_0$  stand for the maximum likelihood estimates (MLEs) of  $\psi_r$  under  $H_1$ ,  $\psi_s$  under  $H_0$ , and  $\psi$  under  $H_0$ , respectively, and  $\hat{\psi}_{r,01} = \arg \max_{\psi_r} f_1(\mathbf{z}, \mathbf{Z}_K; \psi_r, \hat{\psi}_{s0})$ . The Fisher information matrix for the considered detection problem has the following form

$$\mathbf{F}(\psi) = \mathbb{E} \left[ \frac{\partial \ln f_1(\mathbf{z}, \mathbf{Z}_K)}{\partial \psi^*} \frac{\partial \ln f_1(\mathbf{z}, \mathbf{Z}_K)}{\partial \psi^T} \right] \\ = \begin{bmatrix} \mathbf{F}_{\psi_r, \psi_r}(\psi) & \mathbf{F}_{\psi_r, \psi_s}(\psi) \\ \mathbf{F}_{\psi_s, \psi_r}(\psi) & \mathbf{F}_{\psi_s, \psi_s}(\psi) \end{bmatrix}, \quad (5)$$

and the Schur complement of  $\mathbf{F}_{\psi_s, \psi_s}(\psi)$  is

$$\left\{ [\mathbf{F}^{-1}(\psi)]_{\psi_r, \psi_r} \right\}^{-1} \\ = \mathbf{F}_{\psi_r, \psi_r}(\psi) - \mathbf{F}_{\psi_r, \psi_s}(\psi) \mathbf{F}_{\psi_s, \psi_s}^{-1}(\psi) \mathbf{F}_{\psi_s, \psi_r}(\psi), \quad (6)$$

Under the stated assumptions, we have  $\psi_r = [\phi^T, \phi^H]^T$ , and  $\psi_s = [\mathbf{p}^T, \mathbf{p}^H, \text{vec}^T(\mathbf{R})]^T$  in SNE.

### A. Gradient test

The joint PDF under  $H_\lambda, \lambda = 0, 1$  is given by

$$f_\lambda(\mathbf{z}, \mathbf{Z}_K) = \frac{\exp[-\mathbf{z}_\lambda^H (\mathbf{R} + \mathbf{p}\mathbf{p}^H)^{-1} \mathbf{z}_\lambda - \text{tr}(\mathbf{R}^{-1} \mathbf{S})]}{\pi^{N(K+1)} |\mathbf{R} + \mathbf{p}\mathbf{p}^H| |\mathbf{R}|^K}, \quad (7)$$

where  $\mathbf{z}_\lambda = \mathbf{z} - \lambda \mathbf{D}\phi$ , and  $\mathbf{S} = \mathbf{Z}_L \mathbf{Z}_L^H$ . Applying the matrix inversion lemma, we further deduce

$$\frac{\partial \ln f_\lambda(\mathbf{z}, \mathbf{Z}_K)}{\partial \phi^T} = \mathbf{z}_\lambda^H (\mathbf{R} + \mathbf{p}\mathbf{p}^H)^{-1} \mathbf{D} \\ = \mathbf{z}_\lambda^H \left( \mathbf{R}^{-1} \mathbf{D} - \frac{\mathbf{R}^{-1} \mathbf{p} \mathbf{p}^H \mathbf{R}^{-1}}{1 + \mathbf{p}^H \mathbf{R}^{-1} \mathbf{p}} \right) \mathbf{D} \quad (8) \\ = \mathbf{z}_\lambda^H \mathbf{R}^{-1} \mathbf{D},$$

where the constraint in (1) has been employed in the third equality. We denote the MLEs of  $\mathbf{p}$  and  $\mathbf{R}$  under  $H_\lambda$  as  $\hat{\mathbf{p}}_\lambda$  and  $\hat{\mathbf{R}}_\lambda$ , respectively. According to the results in [13],  $\hat{\mathbf{R}}_0$  and the MLE of  $\phi$  under  $H_1$  can be written as

$$\hat{\mathbf{R}}_0^{-1} = (K+1) \mathbf{S}_0^{-\frac{1}{2}} \left( \mathbf{I}_N - \frac{\gamma \mathbf{P}_D^\perp \check{\mathbf{z}} \check{\mathbf{z}}^H \mathbf{P}_D^\perp}{1 + \gamma \check{\mathbf{z}}^H \mathbf{P}_D^\perp \check{\mathbf{z}}} \right) \mathbf{S}_0^{-\frac{1}{2}}, \quad (9)$$

and

$$\hat{\phi}_1 = (\mathbf{D}^H \mathbf{S}^{-1} \mathbf{D})^{-1} \mathbf{D}^H \mathbf{S}^{-1} \mathbf{z}, \quad (10)$$

respectively, where  $\mathbf{S}_0 = \mathbf{S} + \mathbf{z} \mathbf{z}^H$ ,  $\gamma = [\mathbf{z}^H \mathbf{R}^{-1} \mathbf{z} - (K+1) \check{\mathbf{z}}^H \mathbf{P}_D^\perp \check{\mathbf{z}}]^{-1} - 1$ ,  $\check{\mathbf{D}} = \mathbf{S}_0^{-\frac{1}{2}} \mathbf{D}$ , and  $\check{\mathbf{z}} = \mathbf{S}_0^{-\frac{1}{2}} \mathbf{z}$ . It is known from (9) that

$$\hat{\mathbf{R}}_0^{-1} \mathbf{D} = (K+1) \mathbf{S}_0^{-1} \mathbf{D}. \quad (11)$$

Plugging (8), (10), and (11) into (3), and then disregarding constants, leads to the gradient test as

$$\begin{aligned}
t_{\text{Gradient}} &= \mathbf{z}^H (\mathbf{S} + \mathbf{z}\mathbf{z}^H)^{-1} \mathbf{D} (\mathbf{D}^H \mathbf{S}^{-1} \mathbf{D})^{-1} \mathbf{D}^H \mathbf{S}^{-1} \mathbf{z} \\
&= \mathbf{z}^H \left( \mathbf{S}^{-1} - \frac{\mathbf{S}^{-1} \mathbf{z} \mathbf{z}^H \mathbf{S}^{-1}}{1 + \mathbf{z}^H \mathbf{S}^{-1} \mathbf{z}} \right) \mathbf{D} (\tilde{\mathbf{D}}^H \tilde{\mathbf{D}})^{-1} \tilde{\mathbf{D}}^H \tilde{\mathbf{z}} \\
&= \tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}} - \frac{\tilde{\mathbf{z}}^H \tilde{\mathbf{z}} \tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}}}{1 + \tilde{\mathbf{z}}^H \tilde{\mathbf{z}}} \\
&= \frac{\tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}}}{1 + \tilde{\mathbf{z}}^H \tilde{\mathbf{z}}},
\end{aligned} \tag{12}$$

where  $\tilde{\mathbf{z}} = \mathbf{S}^{-\frac{1}{2}} \mathbf{z}$ , and  $\tilde{\mathbf{D}} = \mathbf{S}^{-\frac{1}{2}} \mathbf{D}$ . Note that there is a relationship  $t'_{\text{GLRT}} = (t_{\text{Gradient}}^{-1} - 1)^{-1}$  between (12) and a form of the GLRT-based detector  $t'_{\text{GLRT}}$  for the same considered problem, which is given by (8) of [13]. As a result, these two detectors corresponding to the gradient test and GLRT are shown to be equivalent, since  $t'_{\text{GLRT}}$  is a monotonically increasing function of  $t_{\text{Gradient}}$ .

### B. Durbin test

Building on the results in [13],  $f_{\lambda}(\mathbf{z}, \mathbf{Z}_K)$  and  $[\mathbf{F}^{-1}(\psi)]_{\psi_r, \psi_r}$  can be further derived as

$$\begin{aligned}
f_{\lambda}(\mathbf{z}, \mathbf{Z}_K; \hat{\mathbf{p}}_{\lambda}, \hat{\mathbf{R}}_{\lambda}) &= \frac{(K+1)(1 + \tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}})^{K+1}}{K^{K+1} (e\pi)^{N(K+1)} (\tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}}) |\mathbf{S}_{\lambda}|^{K+1}},
\end{aligned} \tag{13}$$

and

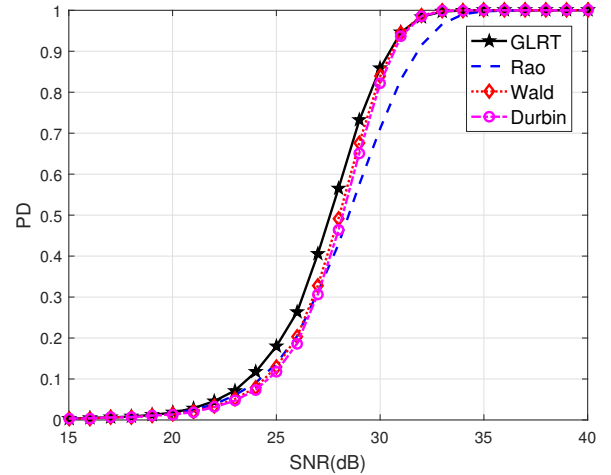
$$\begin{aligned}
[\mathbf{F}^{-1}(\psi)]_{\psi_r, \psi_r} &= [\mathbf{F}_{\psi_r, \psi_r}(\psi)]^{-1} \\
&= [\mathbf{D}^H (\mathbf{R} + \mathbf{p}\mathbf{p}^H)^{-1} \mathbf{D}]^{-1},
\end{aligned} \tag{14}$$

respectively, where  $\mathbf{S}_{\lambda} = \mathbf{S} + \mathbf{z}_{\lambda} \mathbf{z}_{\lambda}^H$ . Taking the derivative of (13) with respect to (w.r.t.)  $\phi$  followed by equating the resulting mathematical expression to zero, yields the estimate  $\hat{\phi}_{r,01} = \arg \max_{\phi_r} f_1(\mathbf{z}, \mathbf{Z}_K; \psi_r, \hat{\psi}_{s0})$ , which has the same expression as  $\hat{\phi}_1$  in (10). Hence, plugging (10), (11), and (14) into (4), and then ignoring constants, we deduce the final Durbin test as

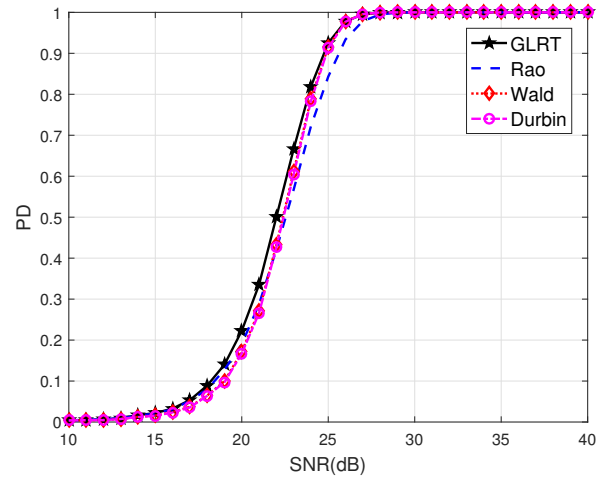
$$\begin{aligned}
t_{\text{Durbin}} &= \tilde{\mathbf{z}}^H \tilde{\mathbf{D}} (\tilde{\mathbf{D}}^H \tilde{\mathbf{D}})^{-1} \mathbf{D}^H (\mathbf{S} + \mathbf{z}\mathbf{z}^H)^{-1} \mathbf{D} (\tilde{\mathbf{D}}^H \tilde{\mathbf{D}})^{-1} \tilde{\mathbf{D}}^H \tilde{\mathbf{z}} \\
&= \tilde{\mathbf{z}}^H \tilde{\mathbf{D}} (\tilde{\mathbf{D}}^H \tilde{\mathbf{D}})^{-1} \mathbf{D}^H \left( \mathbf{S}^{-1} - \frac{\mathbf{S}^{-1} \mathbf{z} \mathbf{z}^H \mathbf{S}^{-1}}{1 + \mathbf{z}^H \mathbf{S}^{-1} \mathbf{z}} \right) \mathbf{D} \\
&\quad \cdot (\tilde{\mathbf{D}}^H \tilde{\mathbf{D}})^{-1} \tilde{\mathbf{D}}^H \tilde{\mathbf{z}} \\
&= \tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}} - \frac{(\tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}})^2}{1 + \tilde{\mathbf{z}}^H \tilde{\mathbf{z}}} \\
&= \frac{\tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}} (1 + \tilde{\mathbf{z}}^H \mathbf{P}_{\tilde{\mathbf{D}}} \tilde{\mathbf{z}})}{1 + \tilde{\mathbf{z}}^H \tilde{\mathbf{z}}}.
\end{aligned} \tag{15}$$

## IV. NUMERICAL EXAMPLES

The Monte Carlo (MC) simulations are utilized to provide numerical examples for performance evaluation of the proposed Durbin test in this section. By comparison, the gradient test coincides with the corresponding GLRT, which has already been assessed in [13]. The detection thresholds, essential for ensuring the predetermined probability of false alarm  $P_{fa} = 10^{-3}$ , are evaluated by exploiting  $10^5$  independent MC trials. In contrast, the probability of detections (PDs) are determined over  $10^4$  independent MC trials. Let  $\text{SNR} = \mathbf{s}^H \mathbf{R}_t^{-1} \mathbf{s}$  be signal-to-noise ratio, and the  $(u, v)$ -th entry of CM  $\mathbf{R}$  is set to be  $0.96^{|u-v|}$ . Moreover, the amount of structural nonhomogeneity, i.e., the amount of the structural mismatch between  $\mathbf{R}_t$  and  $\mathbf{R}$ , is measured by using a quantity  $\rho = \mathbf{p}^H \mathbf{R}^{-1} \mathbf{p}$ . For comparison, the GLRT, Rao test, and Wald test [13] aiming at the hypothesis testing problem under consideration will also be shown as benchmark detectors. The three statistics are given in [13] and omitted here for brevity.



(a)  $K = 2N$



(b)  $K = 4N$

Fig. 1. PD versus SNR for all evaluated detectors when  $N = 72$ ,  $q = 2$ , and  $\rho = 5$  dB.

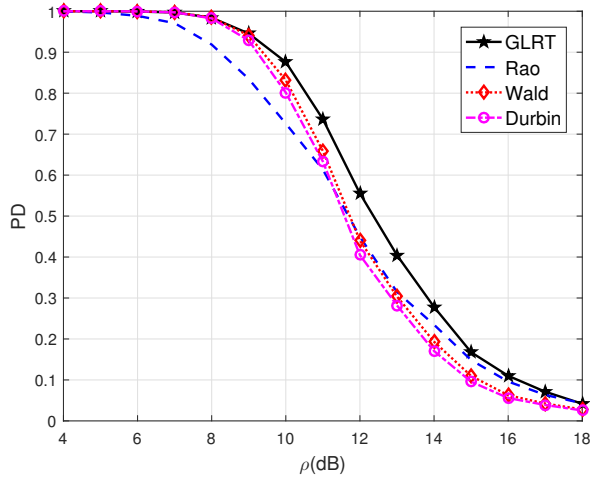


Fig. 2. PD versus  $\rho$  for all evaluated detectors when  $N = 72$ ,  $q = 2$ ,  $K = 2N$ , and  $\text{SNR} = 35$  dB.

Fig. 1 reports the PDs of all evaluated detectors w.r.t. SNR. It is shown that the Durbin test could effectively detect a target for the case of GER-induced structural nonhomogeneity, and the PD of the Durbin test is very close to that of the Wald test. As  $L$  increases, the PDs of the Durbin and Wald tests gradually approach that of the GLRT. The Durbin test exhibits marginal performance degradation in comparison to the GLRT, with SNR losses generally less than 1 dB for  $L = 2N$  and 0.5 dB for  $L = 4N$  under the chosen parameter settings.

With the aim of analyzing the influence of the structural mismatch on detection performance, the PDs of all evaluated detectors at different  $\rho$  levels are displayed in Fig. 2. An increase in  $\rho$  will lead to a greater extent of CM mismatch. When  $\rho$  does not exceed 7 dB, the Durbin test and Wald test achieve the superior performance. As  $\rho$  continues to increase, a marked degradation in PDs can be observed across all detectors. Moreover, the PDs of the Durbin and Wald tests decrease at a similar rate, slightly faster than that of the GLRT. In contrast, even if the Rao test has a relatively slow decreasing PD curve, there is a significant performance decline when  $\rho$  is small.

## V. CONCLUSIONS

This paper reexamined the adaptive detection problem for subspace signals under GER-induced structural nonhomogeneity. The gradient and Durbin tests were designed, and our deduced results show that the Durbin test presents a new detector, whereas the gradient test is coincident with the existing GLRT-based detector. Simulation results confirm that the proposed Durbin test can be capable of effectively detecting a target, and can achieve comparable performance to that of the optimal competitor under the chosen parameter settings. When there is a slight mismatch in the CM structure, the Durbin test performs well. Future research includes the derivation of the statistical performance of the detectors for SNE induced by GER.

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